

Life Cycle Methods

Mounting

Instance of component created and inserted into dom

Updating

A component is re-rendered when its state or props changes.

Unmounting

Occurs when a component is being removed from the DOM

Error Handling

when there is any error during rendering in the lifecycle methods or in the constructor in child components.

ComponentDidMount:

constructor => called when new instance is created, binds event handlers, super.props directly overwrites baseclass constructor(this.state)

static getDerivedStateFromProps => when state of component depends on change in props over time.

As this is static method 'this' cannot use this keyword.

render => read props and state and render the UI.

Do not change state or interact with DOM or make ajax calls.

ComponentDidMount => invoked only once and all child components rendered into DOM.

Causes side Effects, perform any ajax calls to load data.

ComponentDidUpdate:

static getDerivedStateFromProps => rarely used componentDidUpdate
shouldComponentUpdate(nextProps,nextState) = > dictates if component should re render or not.(compares existing and new state and props and return true or false).do not cause sideEffects.

Render() => read this.props and this.state and return jsx

getSnapshotBeforeUpdate(prevProps,prevState) => called before the changes from virtual DOM reflected to the DOM.(this method captures info like prev props and prev state from the dom).Method either return null or a value. Return value is passed as third parameter to the next method.

componentDidUpdate(nextProps,prevProps,snapshot) => called after the render is finished in the re render cycles.cause side Effects.

ComponentWillUnmount:

method invoked before component is destroyed.

Cancelling any network requests,removing event handlers,cancelling any subscriptions and also invalidating timers.

Error Handling Phase methods :

static getDerivedStateFromError(error) => used to render a fallback UI after an error is thrown.

componentDidCatch(error,info) = > used to log the error information

when there is any error either during rendering, in a life cycle method or in the constructor of any child components

Components:

Pure Component : Pure Components implements `shouldComponentUpdate` with a shallow props and state comparison.

Regular Components : Regular components will always implement `shouldComponentUpdate` method it always return true by default.

React.memo():

using **memo function** we can improve the performance of component by not re-rendering the specific Component if its props or state doesnot change.

Example : `export default React.memo(MemoComp);`

React Portals:

React portals provide a way to render children into a DOM node that exists outside the parent hierarchy.